

Ruth Liu

(669) 278-9303 | ruthaliu@gmail.com | www.linkedin.com/in/ruth-liu-b79b25324

EDUCATION

University of California, San Diego

San Diego, CA | September 2023 – June 2027

Bachelor of Science in Mathematics and Computer Science, Minor in Business

- Relevant coursework: Data Structures (Python, Java), Computer Systems (C), Linear Algebra
- CESASC Scholarship Recipient 2025

Notre Dame High School, 3.9 GPA

San Jose, CA | August 2019 – May 2023

- Academic first honors, FIRST Calgames 2021 Innovation Award, FIRST Judges' Award 2022 SVR, Certificate of Merit (CM) Piano Advanced Level

EXPERIENCE

Intern, San Diego Supercomputer Center

San Diego, CA | July 2025 - current

University of California, San Diego

- Developed full-stack React web app enabling campus clubs to manage events and students, implementing responsive UI components and user authentication systems
- Designed and built backend services with [Node.js/Express](#) & MongoDB, implementing RESTful APIs for user management, event CRUD operations, and club membership functionality
- Implemented dynamic club profiles and event management interfaces, using responsive form validation and automated deployment through CI/CD workflows

Event Assistant to Executive Director, UCLA Quantum Innovation Hub

Los Angeles, CA | April-May 2025

University of California, Los Angeles

- Coordinated logistics for a high-profile panel discussion with 100+ attendees, ensuring smooth check-in and setup
- Assisted with operational execution for large-scale event, focusing on efficient event flow

FRC Robotics Team 1967

San Jose, CA | September 2020 - May 2023

Notre Dame High School: Electronics committee member, mini projects leader, YWES Co-host and organizer

- Designed navigation systems and wiring plans for robots in FIRST competitions, contributing to Judges' Award
- Co-hosted Young Women's Engineering Symposium with 40+ attendees and 4 panelists, facilitating STEM workshops
- Mentored 10+ teammates in electronics and system design, designed and prototyped support equipment (robot transport cart, battery cart, custom battery mount), improving team efficiency during competitions

Student VP of Human Resources, Youthpedia Team Lead

San Jose, CA | July 2020 - May 2023

Be the Change Foundation

- Developed standardized orientation session and training materials, improving onboarding procedures for 20+ members
- Led cross-functional collaboration between 5 teams to produce 20+ educational publications, reaching 50+ students

English Instructor

Fremont, CA | July 2023

Fei Teng After School

- Taught 5 English classes to 50+ K-8 students, designing lessons on grammar, creative writing, conversational fluency
- Led project-based learning for recent immigrant students, fostering confidence in public speaking and collaboration

PROJECTS

Amazon Alexa Integration

Palo Alto, CA | June 2022 - August 2022

BlueStamp Engineering

- Developed end-to-end IoT solution using AWS Lambda serverless functions to enable voice-controlled LED lighting and environmental monitoring through Alexa integration

SKILLS

Technical: Python, C, Java, JS, HTML, CSS, Github, Vite + React, Node.js, MongoDB, JUnit, Onshape

Languages: English (Native), Mandarin (Fluent), Cantonese (Fluent)

Certifications: Qiskit Global Summer School 2025 (Quantum Computing fundamentals)